

10. Analysis A: noise vs Z

The noise is split into three parts: temporal (frame-to-frame), fixed-pattern (static, tied to the projector pattern and pixel position), and total. The temporal part is fitted to a disparity-noise model, which yields the forward-model parameters.

1. Load and mask. Convert the depth codes to millimeters with the depth LSB from the SDK (DEPTH_LSB_MM) and set no-reads to NaN. Project the plate outline into the image using the registered pose and shrink it by the 8 px analysis band (BOUNDARY_BAND_HALF_WIDTH_PX) to form the region of interest (ROI).
2. Ground truth. For each ROI pixel, intersect its ray with the registered plate plane. This gives $Z_{GT}(u, v)$.
3. Temporal noise. For each pixel, compute $\sigma_t(u, v)$ as the standard deviation over frames (ddof = 1). Report the ROI median and 95th percentile for each station. Save the σ_t map, since its structure shows the projector-dot layout.
4. Fixed-pattern noise and bias. Average the N frames of the pose to get $\bar{Z}(u, v)$. The mean of $(\bar{Z} - Z_{GT})$ over the ROI is bias(Z). The fixed-pattern noise σ_{fp} is the spread of $\bar{Z} - Z_{GT} - \text{bias}$ over the ROI, measured about the registered plane, with the temporal noise that survives the averaging removed: $\sigma_{fp}^2 = \text{var}(\bar{Z} - Z_{GT}) - \sigma_t^2/N$, where σ_t^2 is the ROI mean of the per-pixel temporal variance (the correction is about 1% of σ_t^2 at 100 frames, and matters only where the fixed pattern is small). Also fit a free plane to $\bar{Z}$ and report the angle between it and the registered plane; the free plane serves only for that angle.
5. Total noise. For each frame, compute the residual from Z_{GT} . σ_{tot} is the RMS over the ROI and frames. Because the three parts are defined about the same registered plane, σ_{tot}^2 should equal $\sigma_t^2 + \sigma_{fp}^2 + \text{bias}^2$ up to sampling error; a mismatch larger than 20% (NOISE_CLOSURE_TOLERANCE) points to an analysis error or to non-stationary noise.
6. Fill rate. Report the fraction of valid pixels in the ROI, for each station and tilt.
7. Spatial correlation. Compute the normalized 2-D autocorrelation of the temporal component ($Z - \bar{Z}$), averaged over frames, and of the fixed-pattern residual map. The correlation length along H and along V is the lag where the normalized autocorrelation first falls to $1/e$, about 0.37. Compare it with the Tier-A IDW neighborhood N and with the 8 px span of the random lateral offsets (PHASE_JITTER_SPAN_PX); the span must exceed it. This is the quick-look that Section 5, Step 7 runs before any B, C, or D capture.

8. Quantization. Histogram the raw depth codes of the central 20×20 px patch (QUANTIZATION_PATCH_PX) over all frames. The most common spacing between populated levels is $\delta Z_q(Z)$. If δZ_q equals the depth LSB at every Z, the output LSB is the quantizer and disparity quantization cannot be seen here; use the B-Z ramp instead. Otherwise compute $q = \delta Z_q \cdot k / Z^2$ at each Z. It should be constant if the quantizer works in disparity.
9. Noise model. Fit $\sigma_t(Z)$ by weighted least squares, each station weighted by the inverse of its σ_t , so that the fit is in relative error and the near stations, where the noise is small, count as much as the far ones. The first term, σ_0 , is a floor that does not depend on Z (LSB/ $\sqrt{12}$ or electronics). The second is disparity noise σ_d propagated through $d = k/Z$. Also fit a free power law $\sigma_t = a \cdot Z^n$. Triangulation predicts $n \approx 2$, and a clearly different n is a finding worth reporting.

$$\sigma_t(Z) = \sqrt{\sigma_0^2 + \left(\frac{\sigma_d Z^2}{k}\right)^2}$$

10. Incidence angle. From the tilt sub-series, plot σ_t , σ_{tot} , and fill rate against incidence angle at each tilt station (Section 5, Step 5; 800 and 1600 mm with the 400 mm plate).
11. Drift correction. Plot the sentinel mean Z against time and sensor temperature, each mounted target relative to its first sentinel after mounting (Section 5, Step 3), and report the drift rate in mm per hour for every mounted target. If a target's drift exceeds a tenth of σ_t at the reference station (WARMUP_DRIFT_FRACTION_OF_SIGMA), subtract a time-interpolated sentinel offset from the bias estimates of the poses captured on that mount, report the size of the correction, and flag the series. If the optional drift run of Section 4, Step 3 was made, fit its depth drift against the sensor temperature reading, predict each sentinel's drift from the logged temperature, and report the difference between the predicted and the observed drift per mounted target; a difference well above the sentinel's own noise points to robot growth or a mount event rather than the sensor.
12. Continuity with earlier data. Also compute the existing testZRepeatabilityBrownBoard.py metrics: in five 20×20 px boxes (half side 10 px, LEGACY_BOX_HALF_PX) at that script's fixed image positions, the mean min–max range and the mean per-pixel standard deviation. They are computed at 700 and 1000 mm (LEGACY_METRIC_DEPTHS_MM), the depths the A series adds to the ladder for this purpose at the center field position. This lets the new results be compared directly with the existing VSX3000 data sets.
13. Outputs. A_noise_summary.csv with one row per station and tilt, carrying the achieved field fraction of the pose (Section 5, Step 1) and the drift rate of the mount it was captured on. Figures (PNG and SVG): the three σ curves vs Z with the model fit; fill rate vs Z; σ_t maps; autocorrelation profiles; depth-code histograms; sentinel drift per mounted target. The fitted σ_d , σ_0 , q, k, and correlation length go in forward_model_parameters.json for the Tier-A simulator.